

eg. Consider the following 3-dimensional dataset with data points classified into three classes: Blue Green and Red

Instance Id	Feature 1 (x_1)	Feature 2 (x_2)	Feature 3 (x_3)	class (y)
A	1	1	1	Blue
B	1	3	2	Blue
C	4	2	5	Green
D	5	3	2	Green
E	6	1	1	Red
F	2	2	2	Red
G	3	3	3	Red
H	4	1	2	Green
I	2	1	3	Red

Classify a new data point with features ($x_1=3, x_2=2, x_3=2$) using K-Nearest Neighbors with $k=3$ using the following distance metrics:- (i) Euclidean (ii) Manhattan (iii) Chebyshev (iv) Cosine Similarity

Sol.New data point, $X = (3, 2, 2)$

Instance Id	Feature 1 (x_1)	Feature 2 (x_2)	Feature 3 (x_3)		$\sqrt{\sum (x_{i1} - x_{i2})^2}$	$\sum x_{i1} - x_{i2} $	$\max x_{i1} - x_{i2} $	$\frac{\sum x_{i1} x_{i2}}{\sqrt{\sum x_{i1}^2} \sqrt{\sum x_{i2}^2}}$
					Euclidean distance	Manhattan Distance	chebyshev Distance	Cosine Similarity
A	1	1	1	(B)	$\sqrt{6}$	4	2	0.980 ←
B	1	3	2	(B)	$\sqrt{5}$	3	2	0.84
C	4	2	5	(G)	$\sqrt{10}$	4	3	0.94
D	5	3	2	(G)	$\sqrt{5}$	3	2	0.984 ←
E	6	1	1	(R)	$\sqrt{11}$	5	3	0.866
F	2	2	2	(R)	$\sqrt{1}$ ←	1 ←	1 ←	0.980 ←
G	3	3	3	(E)	$\sqrt{2}$ ←	2 ←	1 ←	0.980 ←
H	4	1	2	(G)	$\sqrt{2}$ ←	2 ←	1 ←	0.953
I	2	1	3	(R)	$\sqrt{3}$	3	1 ←	0.907

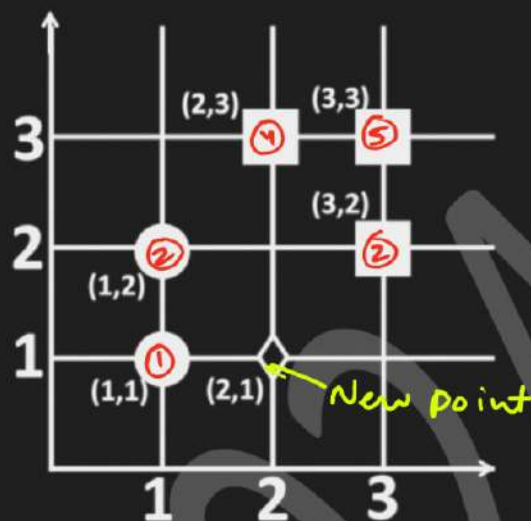
↑
 Smaller is closer
 ↓
 (Red)

↑
 Higher is closer
 ↓
 (Red)

↑
 Higher is closer
 ↓
 (Red)

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Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in k -nearest neighbor algorithm for which the diamond ($\diamond$) shaped data point is assigned the label square is _____.



x_1	x_2	y	(2,1) Euclidean Distance	nearest $k=1$	nearest $k=3$	nearest $k=5$
1	1	C	$\sqrt{(2-1)^2 + (1-1)^2} = \sqrt{1}$	← (C)	←	←
1	2	C	$\sqrt{(2-1)^2 + (1-2)^2} = \sqrt{2}$		←	←
2	3	S	$\sqrt{(2-2)^2 + (1-3)^2} = \sqrt{4}$		(C)	← (S)
3	2	S	$\sqrt{(2-3)^2 + (1-2)^2} = \sqrt{2}$		←	←
3	3	S	$\sqrt{(2-3)^2 + (1-3)^2} = \sqrt{5}$			←

$$\min_k (\text{label assigned} = \text{square}) = \boxed{5}$$

eg: Suppose we have a dataset of news articles with their content tokenized into sets of keywords.

Article ID	x Keywords (Set of Words)	y Category
1	{politics, economy, government}	Politics
2	{stock, economy, market}	Business
3	{government, tax, budget}	Politics
4	{music, album, tour}	Entertainment
5	{bond, finance, investment}	Business

Classify a new article with the keywords {economy, stock, bond} using k-nn classifier for $k=3$.
Use Jaccard index as the distance metric.

Sol. data 1:- $J(x_1, x) = J(\underbrace{\{p, e, g\}}_A, \underbrace{\{e, s, b\}}_B) = \frac{|A \cap B|}{|A \cup B|} = \frac{1}{5} = 0.2$

data 2:- $J(x_2, x) = \frac{2}{4} = \frac{1}{2} = \underline{\underline{0.5}}$

$J(x_3, x) = \frac{0}{6} = 0, \quad J(x_4, x) = \frac{0}{6} = 0, \quad J(x_5, x) = \frac{1}{5} = 0.2$

3 nearest points:- 2, 1, 5

(similarity
metric)

predicted label $\hat{y} = \text{majority}(\text{Business, Politics, Business})$

= Business

eg. Suppose we have a dataset of DNA sequences from different species, each represented by a string of nucleotides of equal length. We need to classify a new DNA sequence "ACCT" using KNN with $k=3$ and Hamming distance

SqID	DNA sequence	Species
1	AACGT	X
2	AAGGT	X
3	AACCT	Y
4	CCCGT	Y
5	AAGTT	Z

Sol. =

SqID	DNA sequence	Species	Hamming Distance from AACCT
1	AACGT	X	2 ←
2	AAGGT	X	3
3	AACCT	Y	1 ←
4	CCCGT	Y	2 ←
5	AAGTT	Z	3

Label, $\hat{y} = \text{majority}(Y, Y, X) = \boxed{Y}$

=x=